Data Flow Diagram

Logical information flow through AI WeatherWise API

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 2

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

Level-1 DFD (textual representation)

Flow	From	To	Data
1	User / API Client	Express API	Credentials, city query, location payload, AI request
2	Express API	MongoDB Atlas	Users and saved locations
3	Express API	OpenWeatherMap	City and weather request
4	OpenWeatherMap	Weather Service	Meteorological JSON
5	Express API	Gemini 2.5 Flash	Weather context + controlled prompt
6	Gemini / fallback	Express API	Summary or recommendation
7	Express API	User / API Client	JSON response + status metadata

Process Logic

- Authenticate protected operations before accessing user-specific location data.
- Weather Service attempts live upstream retrieval and switches to deterministic mock data on failure.
- AI Service attempts Gemini generation and uses rule-based recommendations when quota/network failures occur.
- Controllers return consistent JSON responses so a web, mobile or testing client can consume the same backend.

DFD Symbol / Entity Legend

Entity / Process	Type	Role
User / Thunder Client / Web / Mobile	External entity	Initiates requests and consumes JSON responses.
Express API	Process	Routes, validates and orchestrates requests.
MongoDB Atlas	Data store	Persists users and favorite locations.
OpenWeatherMap	External service	Provides live weather observations.
Gemini 2.5 Flash	External AI service	Produces natural-language contextual intelligence.
Fallback Engine	Local process	Maintains useful responses when external services fail.